



Andrej Karpathy

I like to train deep neural nets on large datasets 🧠 🤖 ⚡



2024 -



I create educational videos on AI on my [YouTube channel](#). The videos come in two parallel tracks: a technical track and a general audience track.

1. Technical track: Follow the [Zero to Hero](#) playlist. General audience track:
2. [Deep Dive into LLMs like ChatGPT](#) is on under-the hood fundamentals of LLMs.
3. [How I use LLMs](#) is a more practical guide to examples of use in my own life.
4. [Intro to Large Language Models](#) is a third, parallel, video from a longer time ago.

For all the latest, I spend most of my time on [X/Twitter](#) or [GitHub](#).

2023 -
2024



I came back to [OpenAI](#) where I built a new team working on midtraining and synthetic data generation.

2017 -
2022



I was the [Director of AI at Tesla](#), where I led the computer vision team of [Tesla Autopilot](#) and (very briefly) [Tesla Optimus](#). My team handled all in-house data labeling, neural network training and deployment on Tesla's custom inference chip. Today, the Autopilot increases the safety and convenience of driving, but the team's goal is to make [Full Self-Driving](#) a reality at scale. See Aug 2021 [Tesla AI Day](#) for more.

2015 -
2017



I was a research scientist and a founding member at [OpenAI](#).

2011 -
2015



My PhD was focused on convolutional/recurrent neural networks and their applications in computer vision, natural language processing and their intersection. My adviser was [Fei-Fei Li](#) at the Stanford Vision Lab and I also had the pleasure to work with [Daphne Koller](#), [Andrew Ng](#), [Sebastian Thrun](#) and [Vladlen Koltun](#) along the way during the first year rotation program.

I designed and was the primary instructor for the first deep learning class Stanford - [CS 231n: Convolutional Neural Networks for Visual Recognition](#). The class became one of the largest at Stanford and has grown from 150 enrolled in 2015 to 330 students in 2016, and 750 students in 2017.

Along the way I squeezed in 3 internships at (baby) Google Brain in 2011 working on learning-scale unsupervised learning from videos, then again in Google Research in 2013 working on large-scale supervised learning on YouTube videos, and finally at DeepMind in 2015 working on the deep reinforcement learning team with [Koray Kavukcuoglu](#) and [Vlad Mnih](#).

2009 -
2011



MSc at the University of British Columbia where I worked with [Michiel van de Panne](#) on learning controllers for physically-simulated figures (i.e., machine-learning for agile robotics but in a physical simulation).

2005 -
2009



BSc at the University of Toronto with a double major in computer science and physics and a minor in math. This is where I first got into deep learning, attending [Geoff Hinton's](#) class and reading groups.

bio

Andrej Karpathy is an AI researcher and educator. He was a founding member of OpenAI and later the Director of AI at Tesla, where he led the computer vision team of the Autopilot. During his PhD at Stanford he was the architect and lead instructor of the first deep learning course at Stanford (CS231n), which has become one of its most popular classes.

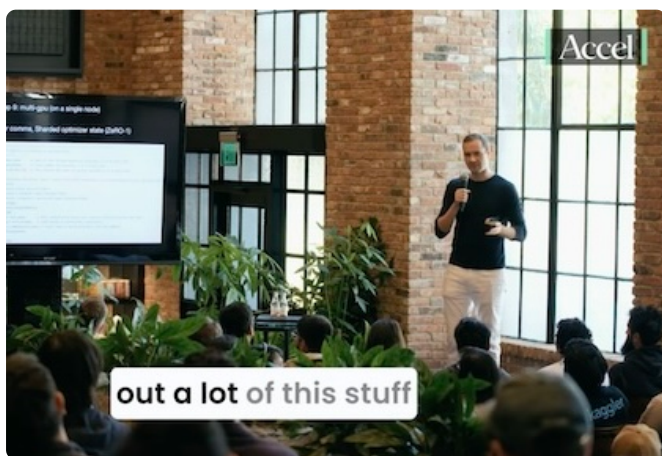
featured talks



Dwarkesh podcast 2025



YC AI Startup School 2025



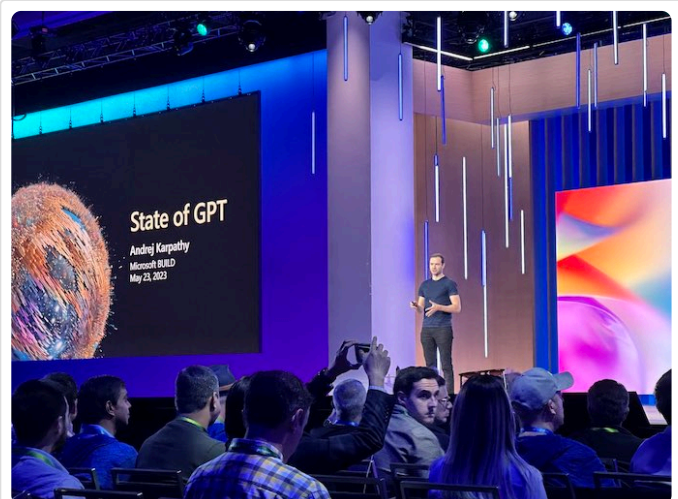
GPU Mode 2024



No Priors podcast 2024



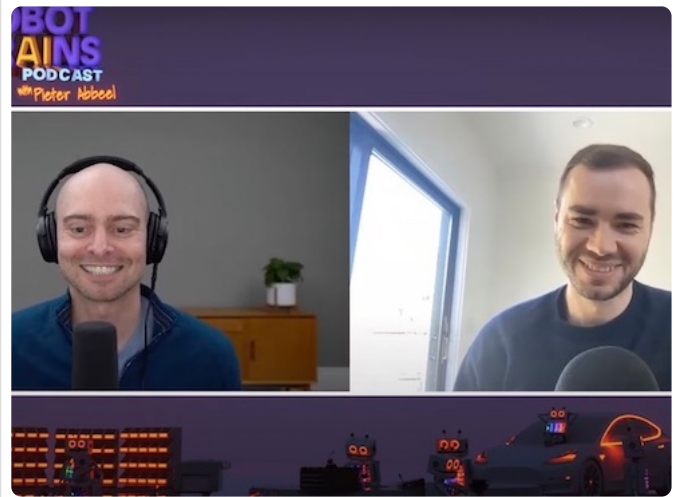
UC Berkeley AI Hackathon 2024



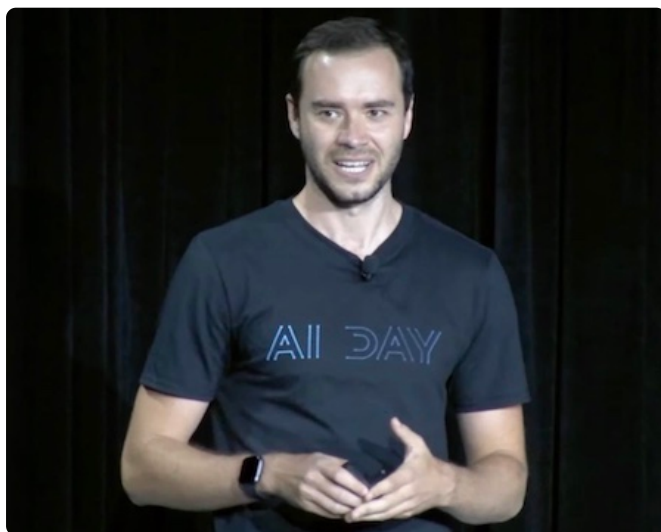
State of GPT @ Microsoft Build 2023 ([slides](#))



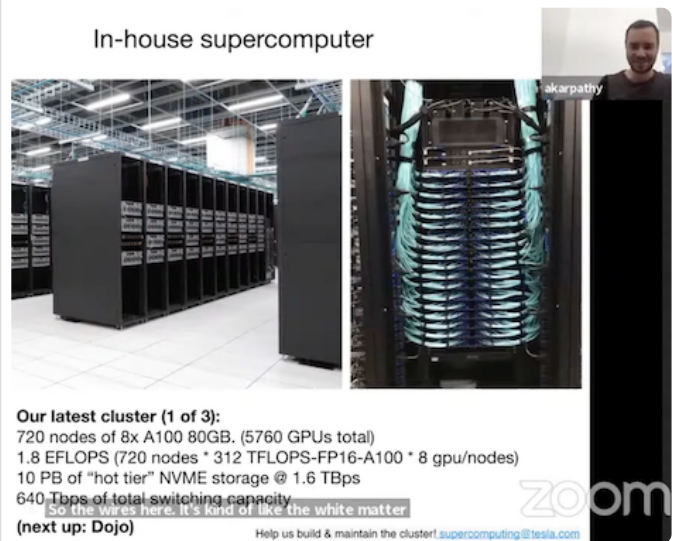
Lex Fridman podcast 2022



Robot Brains podcast with Pieter Abbeel 2021



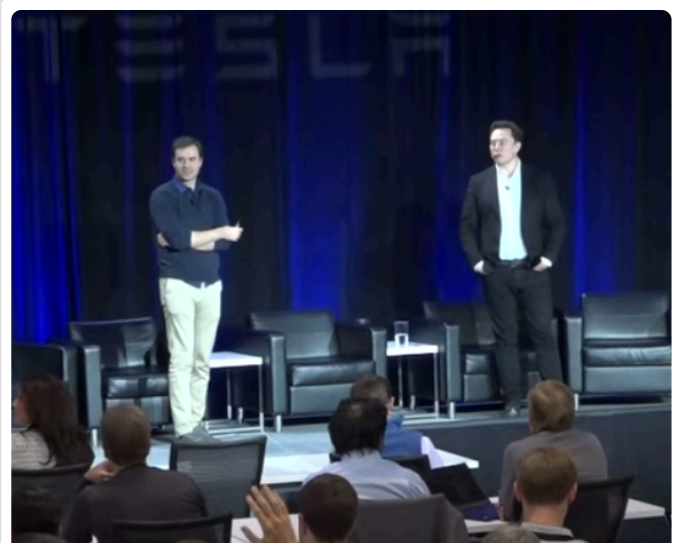
Tesla AI Day 2021



AI for Full Self-Driving @ CVPR 2021



AI for Full Self-Driving @ ScaledML 2020



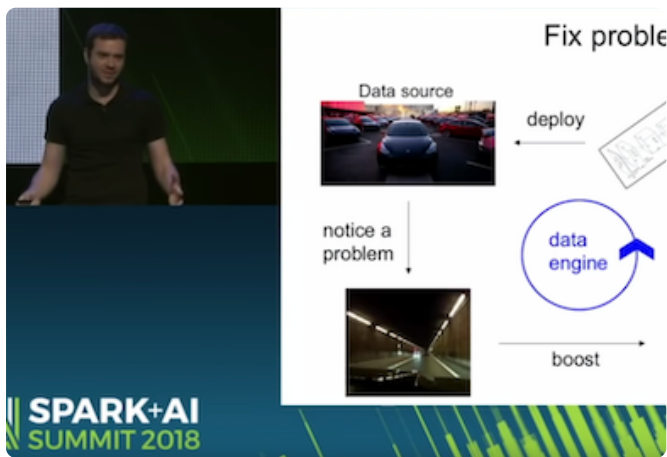
Tesla Autonomy Day 2019



Multi-Task Learning in the Wilderness @ ICML 2019



PyTorch at Tesla @ PyTorch DevCon 2019



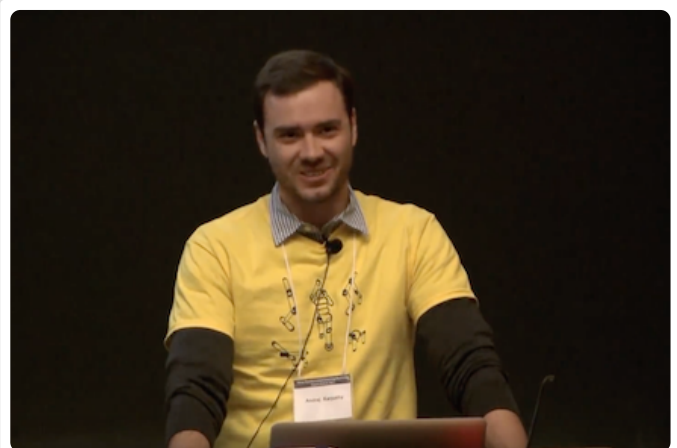
Building the Software 2.0 stack @ Spark-AI 2018



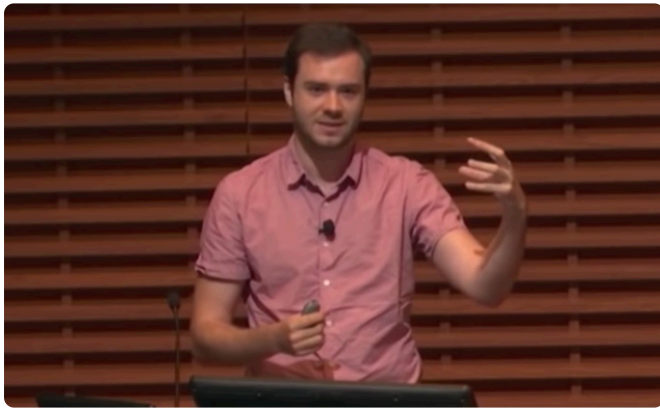
2017 RE•WORK Summit with Nathan Benaich



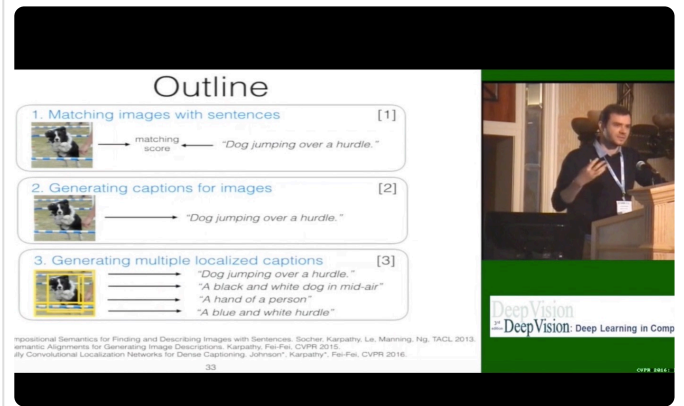
2017 "Heroes of Deep Learning" with Andrew Ng



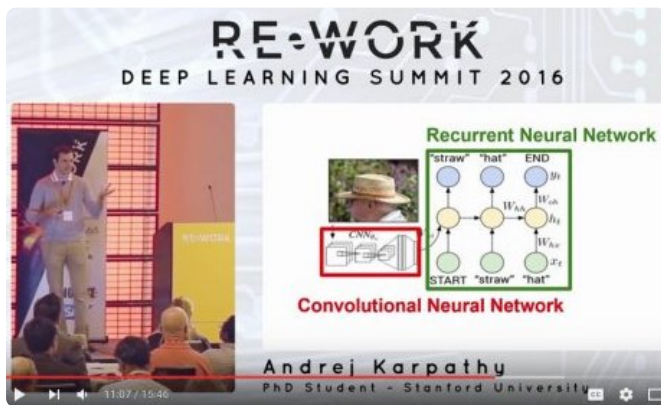
2017 Deep RL Bootcamp with Pieter Abbeel et al



2016 Bay Area Deep Learning School: CNNs



Deep Learning Workshop @ CVPR 2016



RE•WORK Deep Learning Summit 2016



NVIDIA GTC Keynote 2015 with Jensen Huang

teaching

I have a [YouTube channel](#), where I post lectures on LLMs and AI more generally.



[1hr Talk] Intro to Large Language Models

1M views • 4 weeks ago



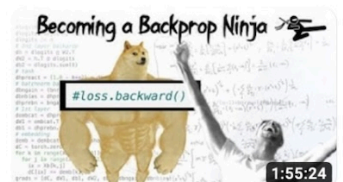
Let's build GPT: from scratch, in code, spelled out.

3.5M views • 11 months ago



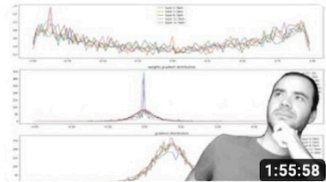
Building makemore Part 5: Building a WaveNet

126K views • 1 year ago



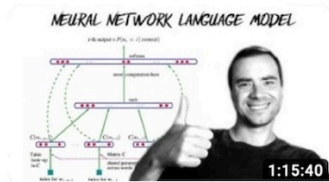
Building makemore Part 4: Becoming a Backprop Ninja

127K views • 1 year ago



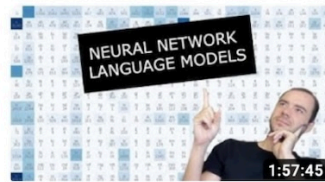
Building makemore Part 3: Activations & Gradients, BatchNorm

184K views • 1 year ago



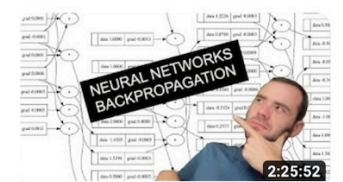
Building makemore Part 2: MLP

210K views • 1 year ago



The spelled-out intro to language modeling: building makemore

440K views • 1 year ago



The spelled-out intro to neural networks and backpropagation...

1.2M views • 1 year ago

In 2015 I designed and was the primary instructor for the first deep learning class Stanford - [CS 231n: Convolutional Neural Networks for Visual Recognition](#) ❤️. The class became one of the largest at Stanford and has grown from 150 enrolled in 2015 to 330 students in 2016, and 750 students in 2017.

[my 2016 lecture videos](#)

[course notes](#)

[course syllabus](#)

[r/cs231n](#)



featured writing

I have three blogs 🏆. This [GitHub blog](#) is my oldest one. I then briefly and sadly switched to my [second blog](#) on Medium. I now have a [Bear blog](#). Here is the (a bit outdated) collection of some of my most popular posts:

Mar 2021 [A from-scratch tour of Bitcoin in Python](#)

Mar 2021 [Short Story on AI: Forward Pass](#)

Jun 2020 [Biohacking Lite](#)

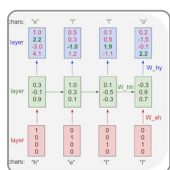
Apr 2019 [A Recipe for Training Neural Networks](#)
Nov 2017 [Software 2.0](#)
Sep 2016 [A Survival Guide to a PhD](#)
Nov 2015 [Short Story on AI: A Cognitive Discontinuity](#)
May 2015 [The Unreasonable Effectiveness of Recurrent Neural Networks](#)
Sep 2014 [What I learned from competing against a ConvNet on ImageNet](#)
Oct 2012 [The state of Computer Vision and AI: we are really, really far away](#)

pet projects

This list is now very outdated :), see my up to date projects on my [GitHub](#).



[micrograd](#) is a tiny scalar-valued autograd engine (with a bite! :)). It implements backpropagation (reverse-mode autodiff) over a dynamically built DAG and a small neural networks library on top of it with a PyTorch-like API.



[char-rnn](#) was a Torch character-level language model built out of LSTMs/GRUs/RNNs. Related to this also see the [Unreasonable Effectiveness of Recurrent Neural Networks](#) blog post, or the [minimal RNN gist](#).



[arxiv-sanity](#) tames the overwhelming flood of papers on Arxiv. It allows researchers to discover relevant papers, search/sort by similarity, see recent/popular papers, and get recommendations. Deployed live at [arxiv-sanity.com](#). My obsession with meta research involved many more projects over the years, e.g. see [pretty NIPS 2020 papers](#), [research lei](#), [scholaroctopus](#), and [biomed-sanity](#). Update: my most recent [arxiv-sanity-lite](#) from-scratch rewrite is much better.

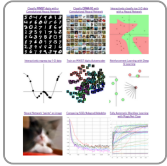


[neuraltalk2](#) was an early image captioning project in (lua)Torch. Also see our later extension with Justin Johnson to [dense captioning](#).

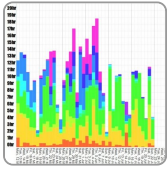


I am sometimes jokingly referred to as the reference human for ImageNet because I competed against an early ConvNet on categorizing images into 1,000 classes. This required a bunch of custom tooling and a lot of learning about dog breeds. See the blog post ["What I learned from competing against a ConvNet on ImageNet"](#). Also a [Wired article](#).

[ConvNetJS](#) is a deep learning library written from scratch entirely in Javascript. This enables nice web-based demos that train convolutional neural networks (or ordinary ones) entirely in the browser. Many web demos included. I did an interview with Data Science Weekly about the



library and some of its back story [here](#). Also see my later followups such as [tSNEJS](#), [REINFORCEjs](#), or [recurrentjs](#), [GANs in JS](#).



How productive were you today? How much code have you written? Where did your time go? For a while I was really into tracking my productivity, and since I didn't like that RescueTime uploads your (very private) computer usage statistics to a cloud I wrote my own, privacy-first, tracker - [ulogme](#)! That was fun.



misc: I built a lot of other random stuff over time. [Rubik's cube color extractor](#), [predator prey neuroevolutionary multiagent simulations](#), [more of those](#), [sketcher bots](#), games for computer game competitions [#1](#), [#2](#), [#3](#), random [computer graphics things](#), [Tetris AI](#), [multiplayer coop tetris](#), etc.

publications

[World of Bits: An Open-Domain Platform for Web-Based Agents](#) ICML 2017

Tianlin (Tim) Shi, Andrej Karpathy, Linxi (Jim) Fan, Jonathan Hernandez, Percy Liang

[PixelCNN++: A PixelCNN Implementation with Discretized Logistic Mixture Likelihood and Other Modifications](#) ICLR 2017

Tim Salimans, Andrej Karpathy, Xi Chen, Diederik P. Kingma, and Yaroslav Bulatov

[Connecting Images and Natural Language \(PhD thesis\)](#) 2016

Andrej Karpathy

[DenseCap: Fully Convolutional Localization Networks for Dense Captioning](#) CVPR 2016 (Oral)

Justin Johnson*, Andrej Karpathy*, Li Fei-Fei

[Visualizing and Understanding Recurrent Networks](#) ICLR 2016 Workshop

Andrej Karpathy*, Justin Johnson*, Li Fei-Fei

[Deep Visual-Semantic Alignments for Generating Image Descriptions](#) CVPR 2015 (Oral)

Andrej Karpathy, Li Fei-Fei

[ImageNet Large Scale Visual Recognition Challenge](#) IJCV 2015

Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, Alexander C. Berg, Li Fei-Fei

[Deep Fragment Embeddings for Bidirectional Image-Sentence Mapping](#) NIPS 2014

Andrej Karpathy, Armand Joulin, Li Fei-Fei

[Large-Scale Video Classification with Convolutional Neural Networks](#) CVPR 2014 (Oral)

Andrej Karpathy, George Toderici, Sanketh Shetty, Thomas Leung, Rahul Sukthankar, Li Fei-Fei

[Grounded Compositional Semantics for Finding and Describing Images with Sentences](#) TACL 2013

Richard Socher, Andrej Karpathy, Quoc V. Le, Christopher D. Manning, Andrew Y. Ng

[Object Discovery in 3D scenes via Shape Analysis](#) ICRA 2013

Andrej Karpathy, Stephen Miller, Li Fei-Fei

[Emergence of Object-Selective Features in Unsupervised Feature Learning](#) NIPS 2012

Adam Coates, Andrej Karpathy, Andrew Ng

[Curriculum Learning for Motor Skills](#) AI 2012

Andrej Karpathy, Michiel van de Panne

[Locomotion Skills for Simulated Quadrupeds](#) SIGGRAPH 2011

Stelian Coros, Andrej Karpathy, Benjamin Jones, Lionel Reveret, Michiel van de Panne

Also on [Google Scholar](#)

misc unsorted

- [Neural Networks: Zero To Hero lecture series](#)
- My [first blog](#) my [second blog](#) and my [current blog](#).
- I like sci-fi. I enumerated and sorted sci-fi books I've read [here](#)
- [Justin Johnson](#) and I held a reading group on Clubhouse. See [YouTube](#) or as [podcast](#).
- [Loss function Tumblr](#) :D! My collection of funny loss functions.
- Some advice for [undergrads](#) and advice for those [considering or pursuing a PhD](#)
- [New York Times article](#) covering my PhD image captioning work.
- t-SNE visualization of [CNN codes for ImageNet](#), pretty!
- A long time ago I was really into Rubik's Cubes. I learned to solve them in about 17 seconds and then, frustrated by lack of learning resources, created [YouTube videos](#) explaining the Speedcubing methods. These went on to become relatively popular. There's also my long dead [cubing page](#). Oh, and a video of me at a [Rubik's cube competition](#) :)
- 0 frameworks were used to make this simple responsive website because I am becoming seriously allergic to [500-pound websites](#). This one is pure HTML and CSS in two static files and that's it.